

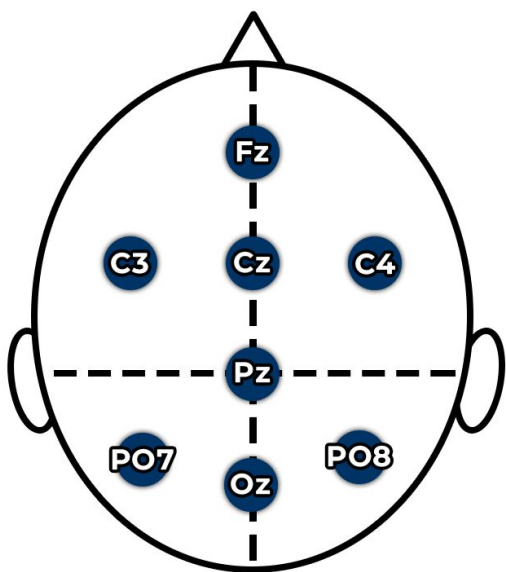
Optimizing EEG Electrode Use for Workload Detection in Noisy and Silent Environments

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Introduction

We tested how reducing EEG electrodes affects workload detection. Using 8 electrodes, participants performed 3-level n-back tasks in silent and noisy environments. Electrode importance for classification was analyzed with a random forest (RF) and a deep neural network (DNN).

How many EEG electrodes are needed to maintain reliable workload classification outside the lab?



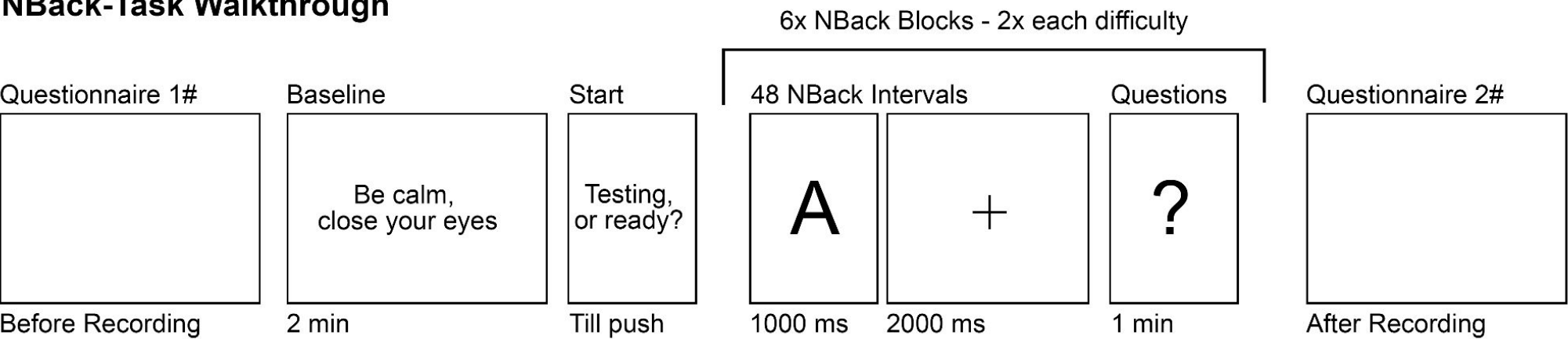
Methods and Data Selection

Subjects	14
Session Files	28
Flawed Sessions	7
Used Sessions	18
Avg. Session Duration	27 min
Total recorded time	654 min

- 14 Subjects, 2 Sessions (Inside/Outside) each
- 3 Difficulty Levels for the NBack-Task
- 8 Electrode EEG Cap, see Fig left
- 1 Baseline Block, 6 Task Blocks (2x each difficulty)
- Microphone recorded noise level
- RF and DNN used for training and classification

Files that were corrupted, could not be loaded, or were missing baseline markers (1 subject) were excluded from the analysis.

NBack-Task Walkthrough

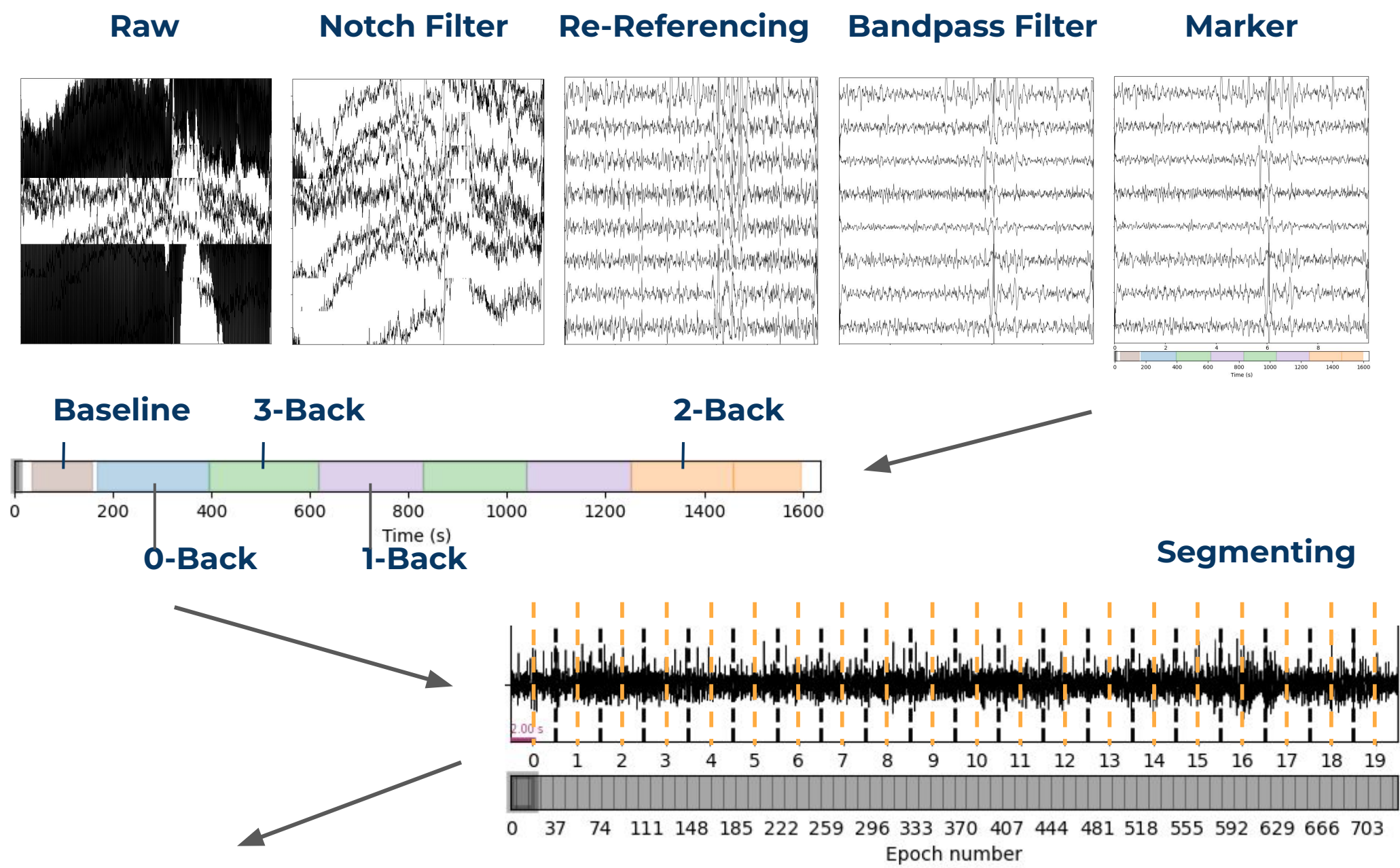


Data Processing

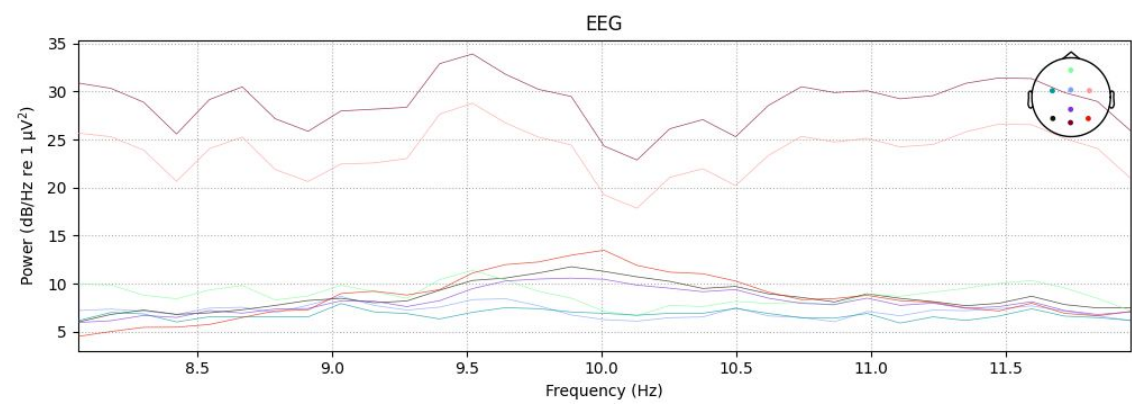
Preprocessing, segmentation and feature extraction

Preprocessing

We applied a 50 and 100 Hz Notch Filter, re-referenced the data with the average of all channels and used a bandpass filter from 4 - 40 Hz. Afterwards we added markers to the EEG-data and segmented the different blocks into 4 sec slices with a 50% overlap.



PSD for Feature Extraction



Feature Extraction

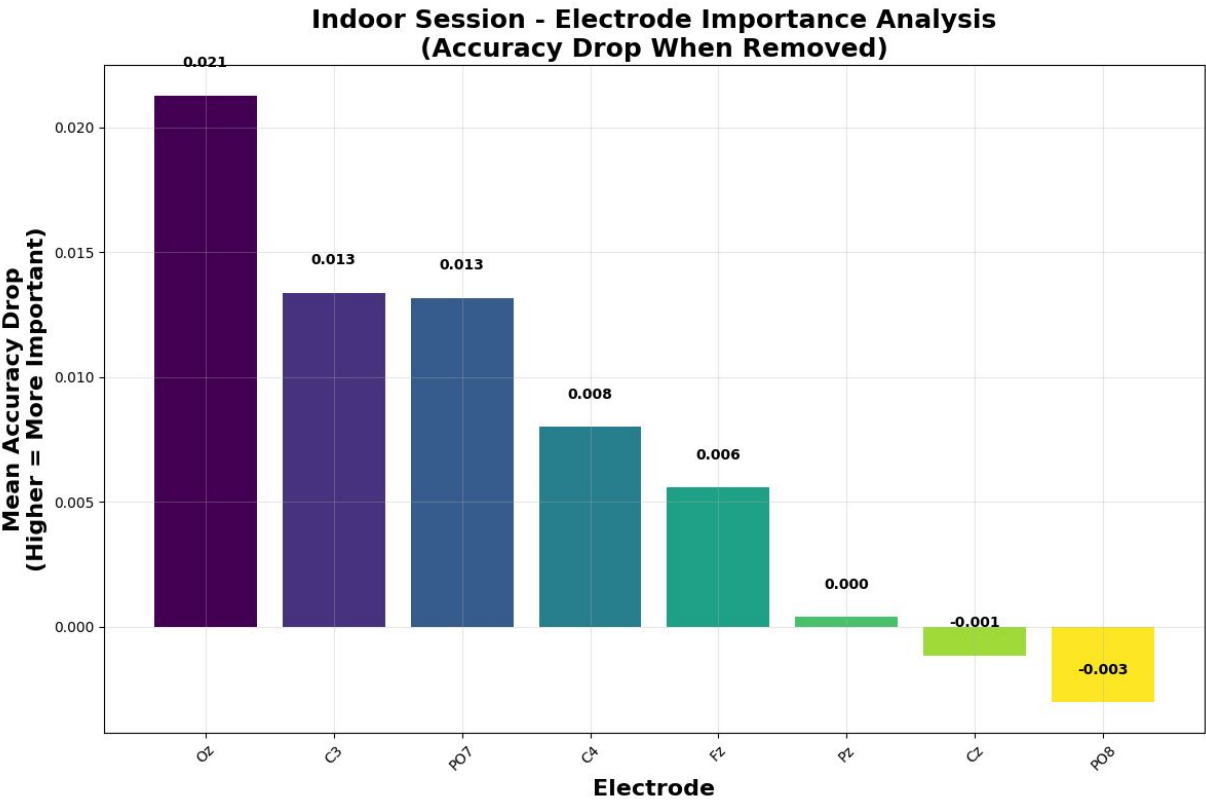
For each electrode and epoch, we sum PSD within $\theta(4-8)$, $\alpha(8-13)$, $\beta(13-30)$, $\gamma(30-40)$ and normalize by total 4 - 40 Hz power → four relative band-power features per electrode.

Time	Condition	Epoch	Fz_ThetaPower	Fz_AlphaPower	Fz_BetaPower	Fz_GammaPower	...
0.00	Baseline	0	-0.34	-1.48	0.0019	2.0085	...
0.004	Baseline	0	-4.38	-0.044	-2.009	2.36	...
...	...	...	...	...	...	...	...

Feature-based ML: Random Forest

Electrode selection and model training

Electrode Selection



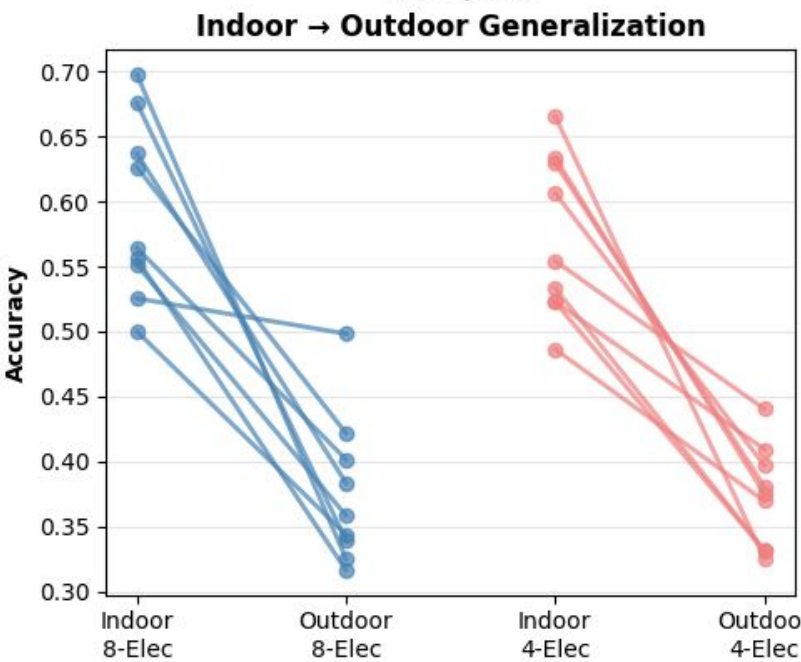
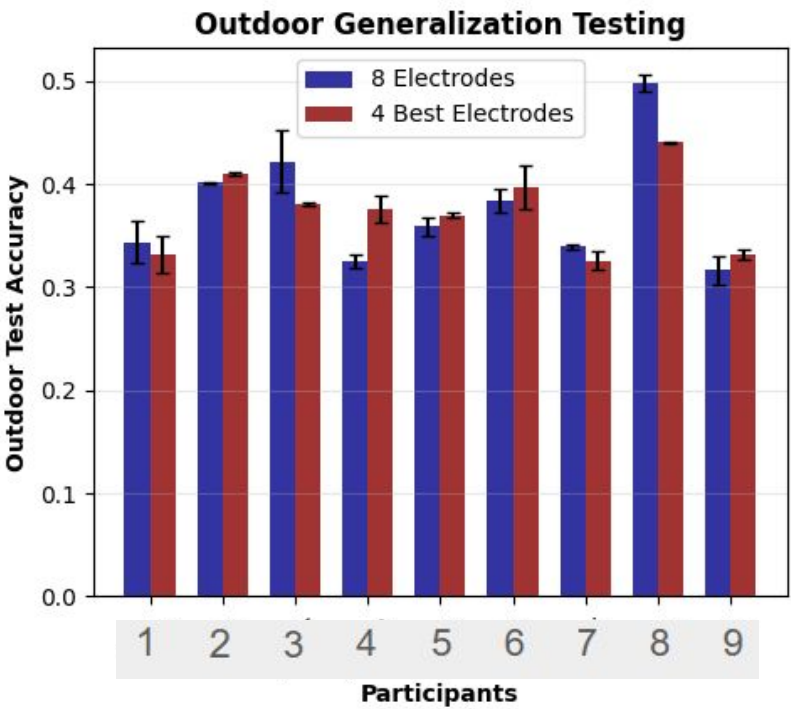
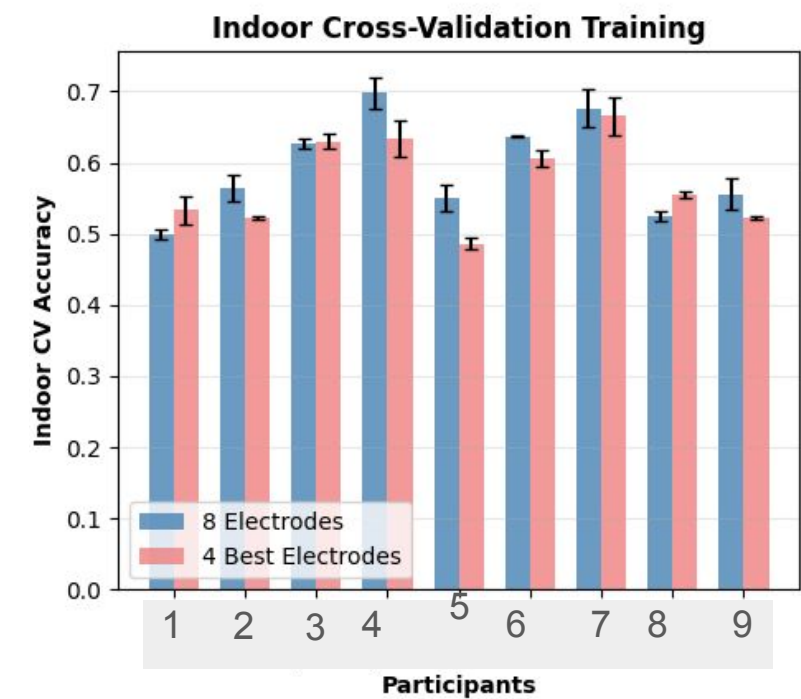
20 best features were selected using the f-values from `classif_f` (sklearn).

Relevant Electrodes were selected by “leave one out”-principle. Electrodes causing the highest accuracy drop when left out were deemed most relevant.

Based on this principle all electrodes were ranked across all subjects to determine the overall best electrodes.

8 vs 4 Electrodes

The model was trained and evaluated using 5-fold cross-validation and compared in classification accuracy to the subset of 4 electrodes (fig.1). With an average classification accuracy of 0.593 for 8 electrodes and 0.573 for 4 electrodes, the subset of 4 electrodes shows a fairly similar result.



Generalization to the outdoor condition showed that the model is not applicable to the outdoor condition in both electrode configurations (fig.2 & fig.3).

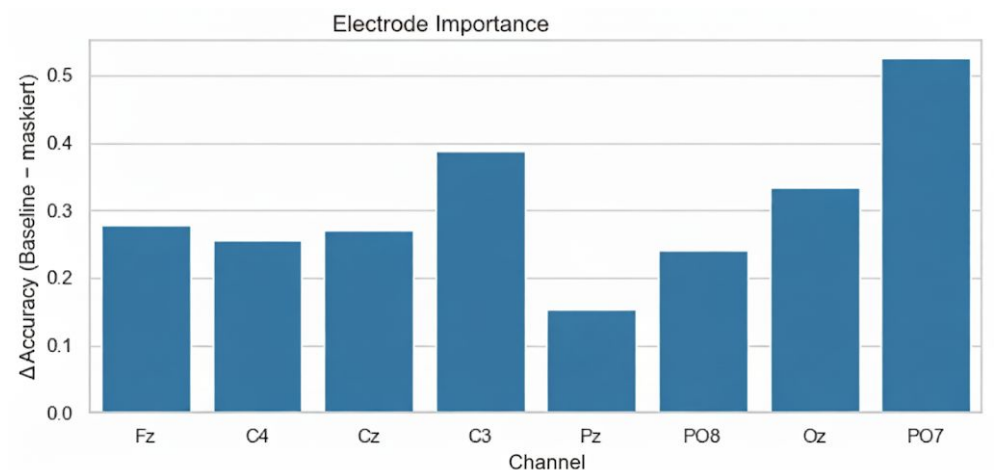
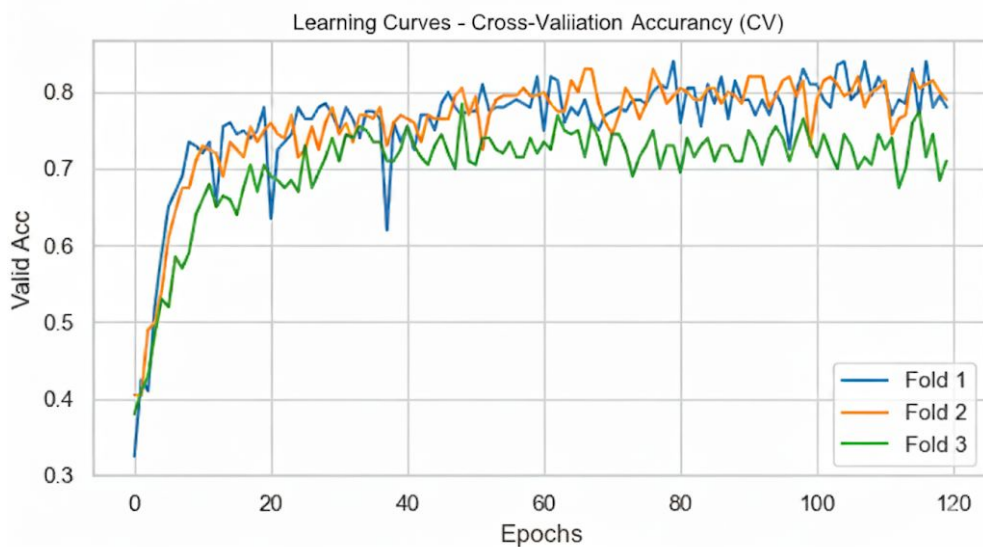
Workload classification can be maintained with as few as four EEG electrodes and only a minor performance decrease.

Deep Learning and Conclusion

Deep Learning, Indoor/Outdoor Transfer and Outlook

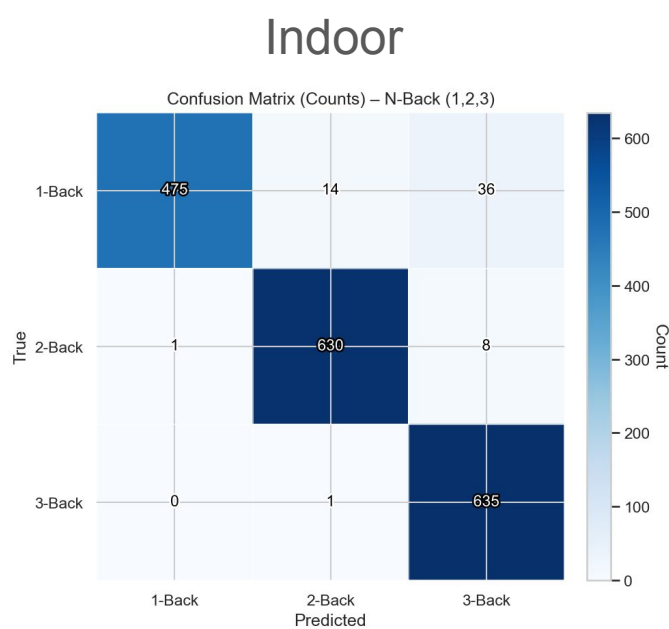
Deep Learning

ShallowFBCSPNet, EEGNet and AttentionBaseNet from braindecode

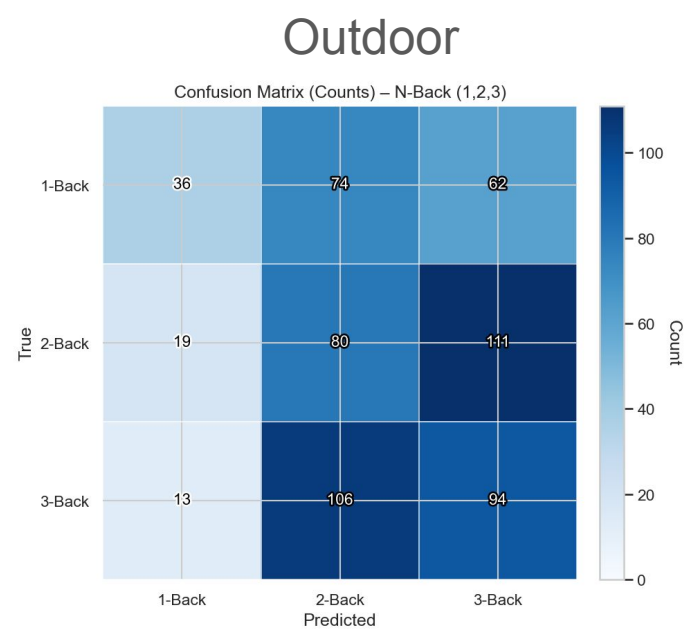


AttentionBaseNet: Normalization (baseline data) + StratifiedKFold + Data Augmentation

Deep Learning - Indoor to Outdoor Transfer



Accuracy: 0.97



Accuracy: 0.35

Summary and Outlook

Reducing from 8 to 4 electrodes caused only a minimal performance drop, which can enable lighter, faster-to-setup real-time BCIs. DNN on the other Hand, profit more from the amount of electrodes, compared to the Random Forest Model.

Data Leakage was observed during training.

Segmentation was applied in the pipeline before the K-Fold cross-validation

Methodological issues in study design (left- and right-handed participants, unstandardized fixation point)

Including the FZ electrode could result in higher accuracy.